Amendment 2 — the endogenous PENK background

Supersedes: two statements in `ENGPENK_AMENDMENT1_WETLAB.md` (see §0). Everything else in Amendment 1, and in `ENGPENK_PEPTIDE_WETLAB.md`, stands. **Basis:** UniProt P01210 (proenkephalin-A, 267 aa), analysed against the engineered insert. **Date:** 2026-08-19. **Status:** dry-lab finding. Nothing here has been measured in-house.

0. Two corrections to Amendment 1

0.1 — "Native PENK cores end in Met or Leu and are therefore not PAM substrates." True for the enkephalins, **wrong as a general statement**. Core 5 (`YGGFMRRVG`…) is trimmed by CPE and amidated by PAM to **adrenorphin / metorphamide,** `YGGFMRRV`**-NH₂, [M+H]⁺ 984.5196**. Native PENK does make an amidated product.

This cuts in the construct's favour. It is direct evidence that PAM is active in these granules, on this precursor. The predicted amidation of the engineered `…VVG` cassette is on firmer ground than Amendment 1 claimed, not weaker.

0.2 — "Cassettes 4 and 7 will not give clean product." Too harsh. Positions 4 and 7 are exactly where **native** PENK deliberately makes extended peptides — the octapeptide `YGGFMRGL` and the heptapeptide `YGGFMRF` come from those same monobasic flanks. The engineered construct is reproducing native architecture, not failing.

What is genuinely open: the engineered extended species `YGGFEVVGRGL` and `YGGFEVVGRF` are not natural ligands and have no published pharmacology. Characterise them; do not assume they need to be engineered away.

1. The finding that changes the project's risk profile

Enkephalins are cleared in seconds, principally by **neprilysin (NEP)** — the same enzyme the project's siRNA arm targets — and by aminopeptidase N. Counting susceptible bonds:

Liability	**Met-enkephalin (native)**	YGGFEVVG	YGGFEVV **-NH₂**
Aminopeptidase, Tyr1-Gly2	1	1	1
NEP internal sites (P1′ hydrophobic)	**1** (Gly3-Phe4)	**3** (Gly3-Phe4, Glu5-Val6, Val6-Val7)	**2**
ACE (C-terminus open)	yes	yes	no
Met oxidation satellite	1	**0**	**0**

The engineered peptide is less protease-stable than native Met-enkephalin, not more. The two valines add NEP-susceptible bonds.

This is where the deltorphin analogy breaks. Deltorphin II is `Y`**-D-Ala-**`FEVVG`-NH₂, and its metabolic stability comes principally from the **D-Ala at position 2** blocking aminopeptidase attack at Tyr1-Xaa2. A ribosomally expressed peptide is all-L by construction and **cannot reproduce that**. The cassette inherits deltorphin's δ-selectivity motif and none of its protease resistance.

> **Consequence for the project plan.** The NEP siRNA is not an adjunct to the peptide arm — with 3 NEP sites against Met-enkephalin's 1, the knockdown is what makes the engineered peptide viable at all. **The two arms are coupled and must be evaluated together.** A peptide-only experiment will under-read the construct; a knockdown-only experiment will not show the peptide's benefit.

The one clear win for the engineered design: **no Met anywhere**, so no +16 Da oxidation satellite on any construct product. That simplifies quantification and formulation stability.

2. What an untransfected cell already secretes

Native proenkephalin-A carries **7 opioid cores** (6 × `YGGFM`, 1 × `YGGFL`) at precursor positions 100, 107, 136, 186, 210, 230, 261. All have a dibasic site upstream; the downstream flanks differ, and that is what generates the native product set.

Core	Pos	Downstream flank	Released product
1	100	KR	`YGGFM` Met-enkephalin
2	107	KK	`YGGFM` Met-enkephalin
3	136	KK	`YGGFM` Met-enkephalin
4	186	**RGL**KR	`YGGFMRGL` Met-enk-Arg-Gly-Leu
5	210	**RRV**GRPE…	`YGGFMRRV`-NH_2 adrenorphin, via BAM-22 → BAM-20
6	230	KR	`YGGFL` Leu-enkephalin
7	261	**RF** (precursor C-term)	`YGGFMRF` Met-enk-Arg-Phe

Any cell line with an intact PENK locus secretes all of this before you transfect anything. Treat it as assay background, not as construct output.

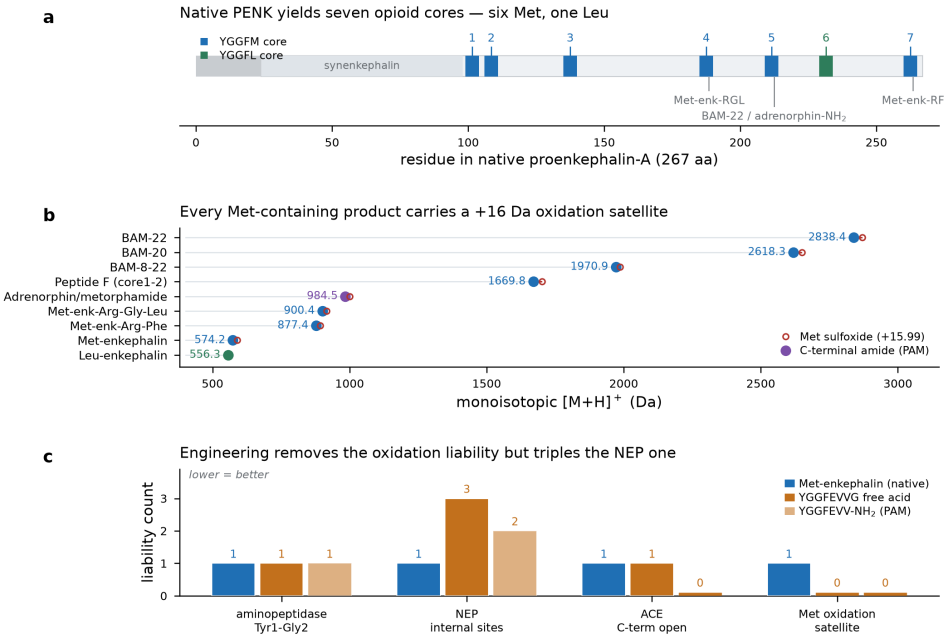


Figure 1. (a) Native proenkephalin-A (267 aa) with its seven opioid cores; cores 4, 5 and 7 carry the monobasic/extended flanks that generate the octapeptide, the BAM series and the heptapeptide — the same architecture the engineered cassettes 4 and 7 reproduce. (b) Monoisotopic $[M+H]^+$ of every native product, with the +15.99 Da Met-sulfoxide satellite drawn as an open circle; BAM-22 and BAM-20 carry two Met. Adrenorphin (purple) is the amidated native product that corrects §0.1. (c) Protease and oxidation liabilities per peptide, counted from sequence. All values computed from UniProt P01210 and the annotated engineered insert.

3. MS target list — 9 endogenous peptides, 20 peaks

6 of 7 native cores carry Met, and Met oxidises to the sulfoxide (**+15.9949 Da**) during extraction, storage and ionisation. Every Met-containing product therefore appears as a doublet — BAM-22 and BAM-20 carry two Met and give a triplet.

Peptide	Sequence	$[M+H]^+$	+1 ox	+2 ox	Origin
`Met-enkephalin`	`YGGFM`	574.2330	590.2279	—	cores 1,2,3 direct; core 5 after full BAM processing
`Leu-enkephalin`	`YGGFL`	556.2766	—	—	core 6

Peptide	Sequence	[M+H]⁺	+1 ox	+2 ox	Origin
`Met-enk-Arg-Gly-Leu`	`YGGFMRGL`	900.4396	916.4345	—	core 4, monobasic downstream
`Met-enk-Arg-Phe`	`YGGFMRF`	877.4025	893.3974	—	core 7, precursor C-terminus
`Adrenorphin/metorphamide`	`YGGFMRRV`	984.5196	1000.5145	—	core 5, PAM product of …RRVG
`BAM-22`	`YGGFMRRVGRPEWWMDYQKRYG`	2838.3504	2854.3453	2870.3402	core 5 intermediate
`BAM-20`	`YGGFMRRVGRPEWWMDYQKR`	2618.2656	2634.2605	2650.2554	core 5 intermediate
`BAM-8-22`	`VGRPEWWMDYQKRYG`	1970.9330	1986.9279	—	core 5, N-terminally trimmed (non-opioid)
`Peptide F (core1-2)`	`YGGFMKRYGGFMKK`	1669.8341	1685.8290	1701.8240	cores 1+2 unseparated intermediate

Full list including the 5 construct species: `penk_ms_target_list.csv`.

Collision check (computed). Across all 25 species — 20 endogenous + 5 construct — there are **no monoisotopic collisions within 1.5 Da between any native and any construct peak**. The two panels are mass-resolved from each other; a quadrupole or ion trap can separate construct output from endogenous background. The only near-isobaric pair in the whole panel is `YGGFEVVG` free acid (827.3934) against `YGGFEVVG`-NH_2 (826.4094), and those belong to two different construct versions that cannot co-occur in one sample.

4. What to run

1. **Untransfected control, same cell line and passage, every time.** Not optional. It defines the endogenous baseline for all 20 peaks above.
2. **Empty-vector control** alongside it, to separate transfection stress from construct output.
3. **Acquire the full 550–2900 Da range**, not a narrow window around the octapeptide. The BAM series sits at 2618–2870 and will be missed by a method tuned to the enkephalins.
4. **Include the +16 satellites in the target list**, or Met-enkephalin will appear to be lost when it has only oxidised.
5. **Report construct peptide as a ratio to endogenous Met-enkephalin** in the same sample. That internally controls for extraction efficiency and cell number, which absolute quantification does not.

5. Sample handling — because 6 of 7 native peaks oxidise

Met oxidation is an artefact of handling as much as of biology. To keep the +16 satellite from becoming the dominant peak:

- Collect on ice; process without a freeze-thaw where possible.
- Acidify immediately (0.1 % formic acid) and keep samples cold and dark.
- Avoid metal surfaces and peroxide-containing solvents; use fresh, degassed mobile phase.
- Run oxidation-prone samples early in the sequence, not after a long queue.
- **Quantify parent + sulfoxide together** as one pool for each Met peptide. This is the robust readout — the parent alone drifts with handling.

6. Bench checklist

- ☐ Untransfected and empty-vector controls scheduled in the same cell line and passage (§4)
- ☐ MS method covers 550–2900 Da, not just the enkephalin window (§4)
- ☐ All 20 endogenous peaks plus their +16 satellites in the target list (§3)
- ☐ Construct peptide reported as a ratio to endogenous Met-enkephalin (§4)
- ☐ Cold/acidified/no-freeze-thaw handling in the SOP (§5)
- ☐ Met peptides quantified as parent + sulfoxide pool (§5)
- ☐ **Project-level:** peptide arm and NEP-siRNA arm evaluated together, not separately (§1)